Lyapunov Stability under Cubic Veronese Lifting

A mathematical guide to the Lean formalization of shadowing, effective cubicization, and fixed-resource transfer

Formalization project companion

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Abstract

This document explains the thirteen LEAN modules that extend the original formalization of fixed-resource homogeneous global asymptotic stability. The extension proves the Lyapunov-stability half of the cubic Veronese reduction. For each fixed source dimension $n \geq 2$ and each fixed $k \geq 1$, one primitive-recursive map sends raw rational codes for degree-$2k+1$ homogeneous fields to raw rational codes for cubic fields in dimension $\binom{n+k-1}{k}$. The same map preserves and reflects both global asymptotic stability and Lyapunov stability with forward existence. The development also proves the semantic shadowing theorem, degree and dimension padding, and transfer of Lyapunov-stability hardness to every larger fixed dimension and every odd degree at least three.

The formal proof follows the mathematical argument through a bounded angular cocycle, exponential attraction to the Veronese cone, and finite-horizon backward shadowing. It does not formalize Korda's undecidability constructions; their conclusions enter the final transfer theorems as explicit hypotheses. It also does not formalize a compiler uniform in the varying pair (n, k). The orbit-shadowing declaration uses the coordinate sup norm, rather than stating verbatim the manuscript's weighted-Q-norm estimate; the two norms are equivalent in the fixed finite-dimensional space, and the formal estimate is the one used to prove the radial-cocycle bound. These distinctions are part of the theorem statements below, not hidden implementation details.

Contents

1 Purpose, scope, and headline result **2**

2 The semantic language and the homogeneous scaling lemma **3**

2.1 Forward trajectories and Lyapunov stability 3

2.2 Stability becomes a global linear bound 4

3 The angular criterion for Lyapunov stability **4**

4 The stabilized cubic Veronese lift **5**

4.1 A global distance estimate 6

4.2 The cubic lift and its shadowing data 6

5 Finite-horizon shadowing **7**

6 The semantic Lyapunov-stability equivalence **8**

7 From semantic lifting to a total computable compiler **8**

7.1 A fixed-parameter effective gain 8

7.2 Malformed codes and the dimension hypothesis 9

8 Padding and transfer of hardness **9**

9 Paper-to-Lean crosswalk **10**

10 Inventory of the thirteen added modules **12**

11 Exact limitations and nonclaims **13**

12 Reproduction and trust boundary **14**

13 Conclusion **14**

1 Purpose, scope, and headline result

The original companion `docs/hgas_sigma1_complete_formalization_companion.tex` documents the fixed-pair and cofinal Σ_1^0-completeness proof for homogeneous GAS. The present document is a second historical artifact. It covers the later Lyapunov-stability and shadowing extension without rewriting the earlier GAS account. The extension adds thirteen modules; the preceding 824 modules remain unchanged.

The main formal statement concerns two languages of raw rational polynomial field codes. The predicate $\mathrm{HGAS}_{\mathbb{Q}}(n,d)$ requires a valid homogeneous degree-d code whose realized field has a globally asymptotically stable origin. The predicate $\mathrm{HLS}_{\mathbb{Q}}(n,d)$ requires the same syntactic validity and the trajectory-model counterpart of Korda's Lyapunov-stability notion: all sufficiently small initial conditions admit a solution for every forward time, and every such solution stays small. For polynomial fields, standard local uniqueness identifies this notion mathematically with the usual maximal-solution formulation. That bridge is not packaged as a separate Lean theorem. Invalid homogeneity codes are negative instances of both languages.

Put

$$m(n,k) = \#\{\alpha \in \mathbb{N}^n : |\alpha| = k\} = \binom{n+k-1}{k}.$$

The Lean definition is `HGAS.Veronese.cubicLiftCodeDimensionnk`.

Theorem 1.1 (The formalized fixed-parameter cubicization)**.** *Fix $n, k \in \mathbb{N}$ with $n \geq 2$ and $k \geq 1$. There is a computable map*

$$\mathcal{C}_{n,k} : \mathrm{Code}(n) \longrightarrow \mathrm{Code}(m(n,k))$$

such that every raw code F satisfies

$$F \in \mathrm{HGAS}_{\mathbb{Q}}(n, 2k+1) \iff \mathcal{C}_{n,k}(F) \in \mathrm{HGAS}_{\mathbb{Q}}(m(n,k), 3),$$
$$F \in \mathrm{HLS}_{\mathbb{Q}}(n, 2k+1) \iff \mathcal{C}_{n,k}(F) \in \mathrm{HLS}_{\mathbb{Q}}(m(n,k), 3).$$

The map $\mathcal{C}_{n,k}$ is the same in the two equivalences.

The exact endpoint is `HGAS.Propagation.exists_computable_cubicization` in `HGAS/Propagation/CofinalLyapunovReduction.lean`. The explicit map is `HGAS.Veronese.effectiveCubicLSReductionCode`; its stronger implementation theorem proves primitive recursiveness, not merely computability.

The hypothesis $n \geq 2$ belongs to the total raw-code theorem. The semantic equivalence between a valid source field and its stabilized cubic lift has no such hypothesis. The restriction appears because the malformed-input branch uses the two-coordinate unstable cubic field

$$\dot{y}_0 = y_0 y_1^2, \qquad \dot{y}_i = 0 \quad (i \neq 0).$$

Section 7 explains this point precisely.

Corollary 1.2 (Cofinal fixed-resource transfer)**.** *Fix $n_0 \geq 2$ and $k_0 \geq 1$, and put $m_0 = m(n_0, k_0)$. If $\mathrm{HLS}_{\mathbb{Q}}(n_0, 2k_0+1)$ is many-one hard, then $\mathrm{HLS}_{\mathbb{Q}}(N, d)$ is many-one hard for every $N \geq m_0$ and every odd $d \geq 3$. Each target language is therefore undecidable and its complement is not recursively enumerable. If only undecidability of the source is assumed, undecidability still transfers to every such target.*

This corollary is unconditional as a transfer principle. Its application to Korda's source family is conditional in Lean because the external undecidability reduction of [1] is not formalized.

2 The semantic language and the homogeneous scaling lemma

2.1 Forward trajectories and Lyapunov stability

The basic dynamical object is a curve $\gamma : \mathbb{R} \to E$ which starts at x_0 and solves $\dot{\gamma} = F(\gamma)$ on $[0, \infty)$. It is represented by `HGAS.IsForwardTrajectoryFgammax0`. Defining the curve on all of $\mathbb{R}$ is a convenient ambient representation; only its restriction to nonnegative time is constrained.

Definition 2.1 (Lyapunov stability with forward existence)**.** *Let E be a real normed space. The origin is Lyapunov stable with forward existence for $F : E \to E$ if, for every $\varepsilon > 0$, there is $\delta > 0$ such that, whenever $\|x_0\| < \delta$,*

1. *at least one forward trajectory through x_0 exists; and*
2. *every forward trajectory through x_0 satisfies $\|\gamma(t)\| < \varepsilon$ for every $t \geq 0$.*

This is the literal definition `HGAS.LyapunovStableWithExistence`. Quantifying over every forward trajectory prevents stability from depending on a choice of solution. For the polynomial fields used here, local uniqueness also identifies it mathematically with the usual maximal-solution formulation; the development does not introduce a separate maximal-solution object. The explicit

existence clause is essential: a definition quantifying only over already global trajectories would hold vacuously near an initial point whose maximal trajectory blows up.

The code language is

$$\mathrm{HLS}_{\mathbb{Q}}(n,d)(F) \iff \begin{array}{l} F \text{ decodes to a degree-}d \text{ homogeneous field,} \\ \text{the realized field is Lyapunov stable with forward existence.} \end{array}$$

It is `HGAS.HLSQ`. Theorems `HGAS.hlsQ_iff_of_homogeneous` and `HGAS.hlsQ_of_hgasQ` expose, respectively, the semantics of valid codes and the implication HGAS $\Rightarrow$ HLS.

2.2 Stability becomes a global linear bound

Suppose that F is homogeneous of degree $k+1$:

$$F(cx) = c^{k+1}F(x) \qquad (c \in \mathbb{R},\ x \in E).$$

Homogeneity rescales both the state and time. If γ is a trajectory, then the appropriate state multiple of $\gamma(c^k t)$ is another trajectory. This observation turns a local stability radius into a global bound.

Theorem 2.2 (Homogeneous scaling criterion)**.** *For a degree-$k+1$ homogeneous field F, the origin is Lyapunov stable with forward existence if and only if there is $C > 0$ such that every $x_0 \in E$ admits a forward trajectory and every forward trajectory through x_0 satisfies*

$$\|\gamma(t)\| \le C\,\|x_0\| \qquad (t \ge 0).$$

The predicate on the right is `HGAS.HasUniformLinearTrajectoryBound`; the equivalence is `HGAS.lyapunovStableWithExistence_iff_uniformLinearTrajectoryBound`. This is the formal analogue of Korda's Lemma 1.5 [1, Lemma 1.5].

The nontrivial direction chooses a stability radius δ for the unit target ball. For $x_0 \neq 0$, scale x_0 by $c = \delta/(2\,\|x_0\|)$, use stability at cx_0, and scale the resulting trajectory back. This gives $C = 2/\delta$. The proof treats $x_0 = 0$ separately and shows, from stability at every scale, that every trajectory from the origin remains at the origin. The reverse implication uses $\delta = \varepsilon/(2C)$.

3 The angular criterion for Lyapunov stability

Let E now be finite-dimensional and Euclidean. For u on the unit sphere define the radial rate and angular field

$$a_F(u) = \langle u, F(u)\rangle, \qquad Y_F(u) = F(u) - a_F(u)u.$$

The angular flow $\phi_\tau(u)$ remains on the sphere. Its logarithmic radial cocycle is

$$A_F(\tau,u) = \int_0^\tau a_F(\phi_s(u))\,\mathrm{d}s.$$

In the implementation, globally cut-off versions of the angular field give a total ambient flow; the cut-off equals one on the sphere, so the displayed objects have the intended values there.

Define boundedness of the forward angular cocycle by

$$\exists M \in \mathbb{R}\ \forall\, \|u\| = 1\ \forall \tau \ge 0, \quad A_F(\tau,u) \le M.$$

This is `HGAS.HasBoundedEuclideanAngularCocycle`.

Theorem 3.1 (Angular Lyapunov criterion)**.** *Let $F : E \to E$ be C^1, homogeneous of degree $k+1$, and satisfy $F(0) = 0$. Then the origin is Lyapunov stable with forward existence if and only if the forward angular cocycle is bounded uniformly over the unit sphere.*

The exact declaration is `HGAS.lyapunovStableWithExistence_iff_boundedEuclideanAngularCocycle`. It formalizes the Euclidean content of Korda's Lemma 2.2 [1, Lemma 2.2].

The proof records the physical-time change rather than treating it as an informal polar-coordinate calculation. The angular reconstruction is

$$\rho(\tau) = e^{A_F(\tau,u)}\phi_\tau(u),$$

and the physical clock has positive speed $e^{-kA_F(\tau,u)}$. The theorem `HGAS.reconstruction_eq_forwardTrajectory` proves that any forward trajectory from a unit vector, sampled at this clock, equals the reconstruction. A uniform linear trajectory bound therefore gives an upper bound for A_F.

Conversely, if $A_F \le M$, the clock speed is at least $e^{-kM} > 0$ on forward angular time. Hence the clock reaches every nonnegative physical time. Inverting it constructs a global trajectory from each unit vector; the reconstruction gives the bound e^M. Homogeneous rescaling supplies trajectories and the same linear bound from arbitrary initial states. The supporting declarations are `HGAS.exists_forwardTrajectory_of_boundedCocycle` and `HGAS.norm_forwardTrajectory_le_of_boundedCocycle`.

Two transport results make the criterion reusable. The declaration `HGAS.lyapunovStableWithExistence_conjugateField_iff` proves invariance under a continuous linear equivalence. The declarations `HGAS.Veronese.lyapunovStableWithExistence_iff_boundedSourceAngularCocycle` and `HGAS.Veronese.lyapunovStableWithExistence_iff_boundedWeightedAngularCocycle` then express Theorem 3.1 in the source coordinates and in the weighted Veronese coordinates used below.

4 The stabilized cubic Veronese lift

Fix $n \ge 1$ and $k \ge 1$. Let

$$A_k = \{\alpha \in \mathbb{N}^n : |\alpha| = k\}, \qquad E_k = \mathbb{R}^{A_k}, \qquad \nu_k(x) = (x^\alpha)_{\alpha \in A_k}.$$

The weighted quadratic form

$$Q(y) = \sum_{\alpha \in A_k} \binom{k}{\alpha} y_\alpha^2$$

satisfies $Q(\nu_k(x)) = |x|^{2k}$. Let q_r range over the balanced quadratic binomials defining the real Veronese cone, and put

$$W(y) = \sum_r q_r(y)^2.$$

Thus $W = 0$ is the cone and $Q = 1,\ W = 0$ is its normalized core.

4.1 A global distance estimate

The shadowing argument needs more than the set-theoretic description of the cone: it needs a quantitative projection estimate.

Proposition 4.1 (Distance to the normalized cone)**.** *There is $C > 0$ such that, for every $y \in E_k$ with $Q(y) = 1$, there is $p \in E_k$ with $Q(p) = 1$, $W(p) = 0$, and*

$$\|y - p\|_Q^2 \le CW(y).$$

This is `HGAS.Veronese.exists_coneDistanceSq_le_energy`. The proof is elementary and is fully formalized. If a pure coordinate y_{ke_i} is bounded away from zero, an explicit affine chart reconstructs a cone point. A one-token transfer relation controls the chart error inductively, coordinate by coordinate. Normalizing the chart point to $Q = 1$ costs only a fixed factor. Compactness of the weighted unit sphere supplies the complementary case: either some pure coordinate is uniformly large or W itself is uniformly large. The latter alternative permits any fixed normalized cone point. The proof therefore does not assume an unformalized tubular-neighborhood or algebraic error-bound theorem.

4.2 The cubic lift and its shadowing data

Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be homogeneous of degree $2k + 1$. The monomial factorization of $D\nu_k(x)F(x)$ defines a cubic field $G : E_k \to E_k$ such that

$$G(\nu_k(x)) = D\nu_k(x)F(x).$$

The stabilized cubic field is

$$H(y) = G(y) - K\nabla_Q W(y).$$

It agrees with G on the cone. On the weighted unit sphere its angular field has the form

$$Y_H = Y_G - K\nabla_{S_Q} W,$$

and its radial rate is $a_H = a_G - 4KW$.

The existing GAS formalization packages the coercivity, tangency, radial, and gain estimates in `HGAS.Veronese.CubicLiftStabilizationBounds`. The new structure `HGAS.Veronese.CubicLiftShadowingBounds` extends this package with a constant $L \ge 0$ satisfying, on $Q = 1$,

$$\begin{aligned} \|Y_G(y) - Y_G(y')\|_\infty &\le L \|y - y'\|_\infty , \\ |a_G(y) - a_G(y')| &\le L \|y - y'\|_\infty , \end{aligned}$$

and with the stronger gain condition

$$2L + 1 \le Kc - C.$$

Here $c > 0$ is the transverse coercivity constant and C bounds the tangency error. We write

$$\lambda = Kc - C > 0.$$

The theorem `HGAS.Veronese.exists_cubicLiftShadowingBounds` proves that such a package exists for every homogeneous source field. The supporting module `HGAS/Veronese/ShadowingPrelims.lean` proves four analytic facts:

1. the angular field and radial reward are Lipschitz on the compact region $Q \le 1$;
2. the tangential penalty gradient satisfies $\left\|\nabla_{S_Q} W\right\|_Q^2 \le C_W^2 W$ on $Q = 1$;
3. a stabilized orbit starting in $W \le \eta$ remains in that tube and satisfies $W(t) \le W(0)e^{-\lambda t}$; and
4. stabilized and unstabilized angular flows agree on $W = 0$.

5 Finite-horizon shadowing

The central obstacle is that convergence of $W(\phi_t^H(y))$ to zero does not identify one cone orbit which follows the stabilized orbit for all future time. Lyapunov stability, however, needs only a uniform upper bound on every finite cocycle segment. The comparison orbit may therefore depend on the terminal horizon.

Theorem 5.1 (Formal orbit-shadowing statement)**.** *Let b be a shadowing package for a homogeneous degree-$2k+1$ source field, with $k \ge 1$. There is $D \ge 0$ such that, for every y with $Q(y) = 1$ and $W(y) \le \eta$, and every horizon $\tau \ge 0$, there is a point q with $Q(q) = 1$ and $W(q) = 0$ such that*

$$\left\|\phi_t^H(y) - \phi_t^H(q)\right\|_\infty \le D\exp(-\lambda t/2) \qquad (0 \le t \le \tau).$$

This is the literal norm and flow formulation of `HGAS.Veronese.exists_finiteHorizonOrbitShadowing`. Because the stabilized and unstabilized flows agree on the cone, $\phi_t^H(q)$ may equivalently be read as the lifted cone flow $\phi_t^G(q)$.

The proof works backward from the endpoint.

1. Exponential penalty decay and the penalty-gradient bound show that the stabilized trajectory is an approximate Y_G-trajectory whose forcing error is $O(e^{-\lambda t/2})$.
2. Proposition 4.1 gives a cone point p within $O(e^{-\lambda\tau/2})$ of the endpoint $\phi_\tau^H(y)$.
3. Flow p backward for time τ on the invariant cone, and call the result q. Then the cone trajectory from q ends at p.
4. Compare the two trajectories backward on $[0, \tau]$. The L-Lipschitz separation of Y_G competes with transverse decay. The inequality $\lambda \ge 2L + 1$ leaves a strict margin and yields the displayed exponential estimate, uniformly in τ.

The radial reward is L-Lipschitz in the same sup norm, so integration of the orbit estimate gives a horizon-independent cocycle error.

Corollary 5.2 (Formal radial-cocycle shadowing)**.** *There is $C_* \in \mathbb{R}$ such that, under the hypotheses of Theorem 5.1, one may choose q with*

$$A_H(y, \tau) \le A_H(q, \tau) + C_*.$$

The exact declaration is `HGAS.Veronese.exists_finiteHorizonShadowing`. This cocycle inequality, rather than the choice of a particular norm in the orbit estimate, is the fact consumed by the stability proof.

Remark 5.3 (Relation to the manuscript's shadowing proposition)**.** *The manuscript states its orbit estimate in the weighted norm $\|\cdot\|_Q$. The named* Lean *theorem states it in the coordinate sup norm carried by* `RealVeroneseSpace`*. In the fixed finite-dimensional space the two norms are equivalent; the development proves the required norm comparisons internally, and the sup-norm estimate implies a weighted-norm estimate after multiplying D by a fixed constant. No public declaration exports that last restatement verbatim. Thus the formalization proves an equivalent-strength shadowing estimate and the exact cocycle conclusion used in the paper, but not the paper's displayed Q-norm sentence word for word.*

6 The semantic Lyapunov-stability equivalence

The angular criterion reduces Lyapunov stability to uniform upper bounds on radial cocycles. On the normalized cone, the lifted cocycle is exactly k times the source cocycle. Hence boundedness for the stabilized lift implies boundedness for the source by restriction to the positive Veronese sheet.

For the converse, consider any stabilized angular orbit. Before it enters the tube $W \le \eta$, the penalty makes its radial rate nonpositive, so this part cannot increase the cocycle. After entry, the tube is forward invariant and Corollary 5.2 compares the remaining finite segment with a cone segment up to C_*. The cone segment is k times a source segment. A source bound M therefore gives a stabilized bound of the form $\max\{0, kM + C_*\}$.

Theorem 6.1 (Semantic preservation and reflection of stability)**.** *Let F be homogeneous of degree $2k+1$, with $k \ge 1$, and let b be any valid shadowing package. Then the origin is Lyapunov stable with forward existence for F if and only if it is Lyapunov stable with forward existence for the stabilized cubic Veronese lift determined by b.*

The exact theorem is `HGAS.Veronese.sourceLS_iff_stabilizedCubicLS`. It is a semantic theorem about polynomial fields and has no $n \ge 2$ assumption. It is the Lyapunov-stability analogue of the older GAS theorem `HGAS.Veronese.sourceGAS_iff_stabilizedCubicGAS`.

7 From semantic lifting to a total computable compiler

7.1 A fixed-parameter effective gain

The semantic theorem permits any sufficiently large gain. A reduction on finite codes must compute one without asking whether the source is stable. For fixed (n, k), expand a homogeneous source field in the finite basis of coordinate monomial fields. Lipschitz constants add under sums and scale by the absolute value of a coefficient. Consequently there is a constant $B_{n,k}$ such that the needed Lipschitz constant is at most

$$B_{n,k} \|F\|_1 ,$$

where $\|F\|_1$ is the sum of the absolute values of the rational coefficients. This is `HGAS.Veronese.exists_cubicLiftLipschitz_uniform`.

The development combines $B_{n,k}$ with the fixed geometric, tangency, and radial constants. It chooses once and for all a natural number $M_{n,k} > 0$ large enough that

$$K(F) = M_{n,k}(1 + \|F\|_1)$$

satisfies every stabilization and shadowing inequality. The package is `HGAS.Veronese.FixedCubicShadowGainParameters`; its existence is `HGAS.Veronese.nonempty_fixedCubicShadowGainParameters`.

The rational code for $K(F)$ is `HGAS.Veronese.effectiveCubicShadowGainCode`. For each fixed (n, k), `HGAS.Veronese.effectiveCubicShadowGainCode_primrec` proves that it is primitive recursive in the input code. The theorem `HGAS.Veronese.exists_effectiveCubicLiftShadowingBounds` connects the decoded gain to the analytic shadowing package.

Warning 7.1 (Fixed-parameter effectivity is not uniform effectivity)**.** *The basis Lipschitz constants and then $M_{n,k}$ are selected using `Classical.choice`. Once n and k are fixed, $M_{n,k}$ is a constant, so hard-coding it yields a genuine primitive-recursive map of F. The development does not give an executable procedure which receives (n, k) as variable input and computes $M_{n,k}$. Therefore it formalizes the cubicization separately for every fixed pair, but it does not formalize the manuscript's uniformity proposition.*

7.2 Malformed codes and the dimension hypothesis

The total compiler `HGAS.Veronese.effectiveCubicLSReductionCode` first tests whether the decoded source is homogeneous of degree $2k + 1$. A valid code is sent to the stabilized cubic lift with gain $K(F)$. An invalid code is sent to `HGAS.Veronese.unstableCubicCode`, the code for $\dot{y}_0 = y_0 y_1^2$, with all other coordinates constant.

This fallback is homogeneous cubic and is neither GAS nor Lyapunov stable. Indeed, on a trajectory with $y_1 = a > 0$, the first coordinate solves $\dot{y}_0 = a^2 y_0$ and therefore grows exponentially. The fully formal proof is `HGAS.Veronese.not_lyapunovStableWithExistence_unstableCubicCode`.

The fallback needs two target coordinates. The theorem `HGAS.Veronese.two_le_cubicLiftCodeDimension` obtains this from $n \geq 2$ and $k \geq 1$. This is the sole source of the $n \geq 2$ assumption in the bundled raw-code HLS reduction. A different one-dimensional negative fallback would remove this packaging restriction; no such alternative is part of the present formalization.

The code-level correctness theorems are

`HGAS.Veronese.effectiveCubicLSReductionCode_hgasQ_iff`,
`HGAS.Propagation.effectiveCubicLSReductionCode_hlsQ_iff`.

Together with primitive recursiveness they yield `HGAS.Propagation.hlsQ_manyOneReducible_cubic` and the simultaneous endpoint in Theorem 1.1.

8 Padding and transfer of hardness

Let $H : \mathbb{R}^m \to \mathbb{R}^m$ be cubic, let $p, j \in \mathbb{N}$, and write $z \in \mathbb{R}^p$. Define

$$\widehat{H}(x, z) = (|x|^2 + |z|^2)^j (H(x), -z_1^3, \ldots, -z_p^3).$$

The field $\widehat{H}$ has dimension $m + p$ and degree $3 + 2j$.

The proof that padding preserves Lyapunov stability has three formal pieces. First, `HGAS.lyapunovStableWithExistence_productField_iff` proves the product criterion. The independent

cubic sinks are GAS, so adjoining them preserves and reflects stability. Second, after a Euclidean coordinate identification, the radial factor equals one on the unit sphere. The theorem `HGAS.lyapunovStableWithExistence_iff_of_eqOn_sphere` proves that two homogeneous C^1 fields of possibly different degrees which agree on the unit sphere have the same Lyapunov-stability answer. Third, the executable polynomial-code construction is connected to this semantic field.

The resulting code theorem is `HGAS.DegreePaddingCode.degreeDimensionPadCode_hlsQ_iff_of_homogeneous`. Composing it with cubicization gives `HGAS.Propagation.hlsQ_manyOneReducible_cofinalFixedPair`:

$$\mathrm{HLS}_{\mathbb{Q}}(n, 2k+1) \leq_{\mathrm{m}} \mathrm{HLS}_{\mathbb{Q}}(N, d)$$

whenever $n \geq 2$, $k \geq 1$, $N \geq m(n,k)$, and $d \geq 3$ is odd.

The formal predicate `HGAS.ManyOneHardP` says that every recursively enumerable predicate on $\mathbb{N}$ computably many-one reduces to P. The general lemmas `HGAS.ManyOneHard.not_computable` and `HGAS.ManyOneHard.complement_not_re` prove that a many-one hard predicate is undecidable and has a non-r.e. complement. They do not assume that P itself is recursively enumerable. This distinction matters for Lyapunov stability, whose membership in the r.e. sets is not established.

The endpoint `HGAS.Propagation.hlsQ_manyOneHard_cofinal_of_source` takes a proof that one source language is many-one hard and returns hardness, undecidability, and non-r.e. complement for every cofinal target pair. The weaker endpoint `HGAS.Propagation.hlsQ_not_computable_cofinal_of_source` transfers undecidability alone.

Korda's construction supplies the source hypothesis in the manuscript [1, Sections 3–6]. The Lean theorem does not contain that construction. In particular, the formalization does not independently prove that any HLS language is many-one hard; it proves the complete compiler and the exact conditional transfer from such a source.

9 Paper-to-Lean crosswalk

Here, calling a result formalized means that the named declaration, together with the declarations listed in its row, proves the paper result or the stated equivalent formulation. It does not mean that every sentence of the printed proof has been translated line by line.

Paper result	Formal status	Lean declaration and module
Homogeneous scaling lemma (Korda Lemma 1.5)	Equivalent trajectory formulation, including forward existence and a bound on every global forward trajectory. No separate maximal-solution bridge is exported.	`HGAS.lyapunovStableWithExistence_iff_uniformLinearTrajectoryBound`, `HGAS/Stability/KordaLyapunov.lean`.

Paper result	Formal status	Lean declaration and module
Angular criterion for Lyapunov stability (Korda Lemma 2.2)	Equivalent Euclidean cocycle criterion in the same trajectory model; transported to the weighted form by a linear equivalence.	`HGAS.lyapunovStableWithExistence_iff_boundedEuclideanAngularCocycle`, `HGAS/Stability/AngularLyapunov.lean`; and `HGAS.Veronese.lyapunovStableWithExistence_iff_boundedWeightedAngularCocycle`, `HGAS/Veronese/LyapunovAngularCriteria.lean`.
Distance comparison for the Veronese cone	Exact existential quadratic error bound in the weighted metric.	`HGAS.Veronese.exists_coneDistanceSq_le_energy`, `HGAS/Veronese/ConeDistance.lean`.
Lipschitz and tube estimates used by shadowing	Both the paper-style weighted estimates and the sup-norm estimates used by the final shadowing package are formalized; fixed-parameter effectivity uses the latter.	`HGAS.Veronese.exists_lipschitz_cubicLift_angular`, `HGAS/Veronese/ShadowingPrelims.lean`; and `HGAS.Veronese.exists_cubicLiftLipschitz_uniform`, `HGAS/Computability/CubicLiftEffectiveShadowGain.lean`.
Finite-horizon shadowing proposition	Split into an orbit theorem and a radial-cocycle theorem. The named orbit theorem uses the coordinate sup norm; the paper displays the weighted norm.	`HGAS.Veronese.exists_finiteHorizonOrbitShadowing` and `HGAS.Veronese.exists_finiteHorizonShadowing`, `HGAS/Veronese/CubicLiftLyapunovConjugacy.lean`.
Cubicization: Lyapunov-stability equivalence	The semantic theorem has no $n \geq 2$ side condition; the total raw-code compiler does.	`HGAS.Veronese.sourceLS_iff_stabilizedCubicLS`, `HGAS/Veronese/CubicLiftLyapunovConjugacy.lean`; and `HGAS.Propagation.effectiveCubicLSReductionCode_hlsQ_iff`, `HGAS/Propagation/CofinalLyapunovReduction.lean`.
Cubicization: one map for GAS and HLS	Exact fixed-(n,k) code theorem for $n \geq 2$; uniformity in varying (n,k) is not formalized.	`HGAS.Propagation.exists_computable_cubicization`, `HGAS/Propagation/CofinalLyapunovReduction.lean`.
Degree raising and dimension padding	Exact HLS equivalence on valid cubic codes; combined with the older GAS padding result.	`HGAS.DegreePaddingCode.degreeDimensionPadCode_hlsQ_iff_of_homogeneous`, `HGAS/Stability/PaddingLyapunov.lean`.

Paper result	Formal status	Lean declaration and module
Cubic normal form, fixed-parameter parts (i) and (ii)	Part (i) is an exact many-one reduction. Part (ii) is covered by the padding equivalence for homogeneous cubic codes composed with cubicization; bundled HLS cubicization requires $n \geq 2$.	`HGAS.Propagation.hlsQ_manyOneReducible_cubic` and `HGAS.Propagation.hlsQ_manyOneReducible_cofinalFixedPair`, `HGAS/Propagation/CofinalLyapunovReduction.lean`.
Transfer of fixed-resource HLS hardness	Exact conditional transfer theorem. The source-hardness premise is explicit.	`HGAS.Propagation.hlsQ_manyOneHard_cofinal_of_source`, `HGAS/Propagation/CofinalLyapunovReduction.lean`.
Consequence of Korda's main reduction	Conditional only: Korda's construction is an external input, not a Lean theorem in this repository.	Instantiate the preceding endpoint with a term of type `HGAS.ManyOneHard(HGAS.HLSQn0(2*k0+1))`.
Dimension-five undecidability consequence	Conditional undecidability transfer; the dimension-five source theorem is external.	`HGAS.Propagation.hlsQ_not_computable_cofinal_of_source`, `HGAS/Propagation/CofinalLyapunovReduction.lean`.

10 Inventory of the thirteen added modules

Module	Principal contribution
`HGAS/Stability/KordaLyapunov.lean`	Defines stability with forward existence and $\mathrm{HLS}_{\mathbb{Q}}$; proves the homogeneous uniform-linear-bound criterion.
`HGAS/Stability/AngularLyapunov.lean`	Proves the Euclidean angular-cocycle criterion, including the physical-clock reconstruction and existence argument.
`HGAS/Stability/AngularCongruence.lean`	Proves that homogeneous fields agreeing on the unit sphere have identical forward angular cocycles and the same Lyapunov-stability answer.
`HGAS/Stability/LinearConjugacyLyapunov.lean`	Proves invariance of Lyapunov stability with existence under continuous linear coordinate equivalences.
`HGAS/Stability/PaddingLyapunov.lean`	Proves product, cubic-sink, degree-raising, dimension-padding, and executable code equivalences for HLS.
`HGAS/Veronese/LyapunovAngularCriteria.lean`	Transports the angular criterion to the source sphere and the weighted Veronese sphere.

Module	Principal contribution
`HGAS/Veronese/ConeDistance.lean`	Proves the global quadratic distance bound from the weighted sphere to the normalized real Veronese cone.
`HGAS/Veronese/ShadowingPrelims.lean`	Supplies analytic Lipschitz bounds, the penalty-gradient estimate, exponential tube decay, and equality of the two flows on the cone.
`HGAS/Veronese/CubicLiftShadowingBounds.lean`	Defines the stronger stabilization package carrying Lipschitz data and the gain inequality needed for shadowing.
`HGAS/Veronese/CubicLiftLyapunovConjugacy.lean`	Proves existence of shadowing data, finite-horizon shadowing, and semantic preservation and reflection of Lyapunov stability.
`HGAS/Computability/ManyOneHard.lean`	Defines many-one hardness and derives undecidability and non-r.e. complement without assuming r.e. membership of the target.
`HGAS/Computability/CubicLiftEffectiveShadowGain.lean`	Builds the fixed-parameter primitive-recursive gain and total compiler, proves the unstable fallback negative, and establishes code-level GAS and conditional HLS correctness.
`HGAS/Propagation/CofinalLyapunovReduction.lean`	Discharges the semantic premise, combines GAS and HLS in one compiler, adds padding, and proves cofinal hardness and undecidability transfer.

11 Exact limitations and nonclaims

The formalization proves the dependency chain needed for the fixed-parameter results. It deliberately makes the following boundaries visible.

1. **The raw-code bundled theorem assumes $n \geq 2$.** The semantic lift is valid for $n \geq 1$. The extra assumption supports the two-coordinate negative fallback for malformed codes.
2. **Computability is fixed-parameter.** For each fixed (n, k), the compiler is primitive recursive in F. The formalization does not compute the hard-coded multiplier from a variable input (n, k), so it does not establish the paper's uniformity proposition.
3. **The public orbit theorem uses the sup norm.** It gives an equivalent-strength finite-dimensional estimate and proves the required cocycle theorem, but it is not a verbatim export of the paper's displayed weighted-Q-norm inequality.
4. **The maximal-solution bridge is mathematical context, not an exported Lean theorem.** The development represents solutions as global forward trajectories and includes forward existence in the stability predicate. Polynomial local uniqueness makes this equivalent to the customary maximal-solution wording, but the repository does not expose a standalone theorem between those two APIs.

5. **Korda's hardness and undecidability constructions are external.** The formalization proves trajectory-model counterparts of Korda's two analytic lemmas used by the cubicization, but not the register-machine reduction of [1] or the dimension-five supplement [2]. The final transfer declarations take the corresponding hardness or undecidability statement as a hypothesis.
6. **Many-one hardness is not called completeness.** No theorem here shows that $\mathrm{HLS}_{\mathbb{Q}}(n, d)$ is recursively enumerable. The formal conclusion from Korda's source is therefore hardness, undecidability, and a non-r.e. complement, not Σ_1^0-completeness.
7. **Variable-resource normal forms are outside this extension.** The formalization does not cover the manuscript's variable-dimension statements, even-degree conversion, decision procedure for linear fields, or the consequence for GAS of fields $H - \mathrm{id}$.

These limitations do not weaken the fixed-dimension, fixed-degree GAS completeness theorem documented in the earlier companion. They delimit the new HLS and shared-cubicization claims.

12 Reproduction and trust boundary

The repository pins Lean and Mathlib at version 4.30.0. From the repository root containing `lakefile.toml`, a fresh build is obtained with

```
lake update
lake build HGAS
```

The public import root is `HGAS.lean`.

The new endpoints are ordinary theorems, not project axioms. Kernel axiom inspection of the audited endpoints reports only `propext`, `Classical.choice`, and `Quot.sound`. Classical choice is used, among other places, to select fixed analytic constants and the multiplier $M_{n,k}$; as explained in Section 7, this is compatible with a computable map for each fixed pair but does not yield a uniform executable function of the pair.

Lean's kernel checks the elaborated proof terms. Mathlib supplies the underlying finite-dimensional analysis, ODE, integration, polynomial, and computability libraries [3, 4]. The nonroutine external mathematical input needed to instantiate the final HLS hardness transfer is Korda's unformalized source reduction. Identifying its maximal-solution wording with Lean's global-forward-trajectory predicate also uses the standard polynomial existence-and-uniqueness context described in Section 2; that API bridge is not itself exported as a Lean theorem.

13 Conclusion

The extension closes the formal gap between a cubic lift that preserves GAS and one that also preserves Lyapunov stability. Its main new idea is the finite-horizon comparison selected from the endpoint of each orbit segment. This construction converts exponential transverse attraction into a uniform radial-cocycle bound without requiring one cone orbit to shadow forever. The formal proof then connects the semantic argument to a primitive-recursive compiler on raw rational codes and to cofinal fixed-resource transfer.

The resulting release should therefore present two complementary artifacts: the earlier GAS completeness companion, preserved as a record of the original 824-module development, and this companion for the thirteen-module HLS and shared-cubicization extension. A coverage ledger can refer readers to both documents while distinguishing exact statements, equivalent formulations, conditional applications, and results not formalized.

References

[1] M. Korda, *Lyapunov stability of polynomial vector fields is undecidable*, arXiv:2609.22058v2, 2026. https://arxiv.org/abs/2609.22058.

[2] M. Korda, *Supplementary material for "Lyapunov stability of polynomial vector fields is undecidable": dimension five, global asymptotic stability, and the planar case*, 2026. https://homepages.laas.fr/mkorda/arnold_supplement.pdf.

[3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in *Automated Deduction—CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, pp. 625–635. https://doi.org/10.1007/978-3-030-79876-5_37.

[4] The Mathlib Community, *The Lean mathematical library*, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, 2020, pp. 367–381. https://doi.org/10.1145/3372885.3373824.